

# Smart Lender: AI-Powered Loan Eligibility Prediction System

## Project Description

### Description

Smart Lender is an Artificial Intelligence and Machine Learning-based web application designed to predict the loan eligibility of applicants. The system helps banks and financial institutions make faster, smarter, and more accurate loan approval decisions by analyzing applicant information using a trained Machine Learning model.

The application collects details such as Gender, Marital Status, Education, Employment Status, Applicant Income, Co-applicant Income, Loan Amount, Loan Amount Term, Credit History, and Property Area. These details are processed using a Random Forest Classification model, which predicts whether the applicant is eligible for a loan.

The system is developed using Flask, HTML, CSS, Bootstrap, JavaScript, SQLite, and Python. It provides secure user authentication, instant prediction results, and stores prediction history for future reference. Smart Lender minimizes manual work, reduces loan approval time, and improves decision-making accuracy.

## Scenarios

### Scenario 1 – Loan Approval

A bank employee enters the details of a salaried applicant with a good credit history. The system predicts that the applicant is eligible for the loan within seconds.

### Scenario 2 – Loan Rejection

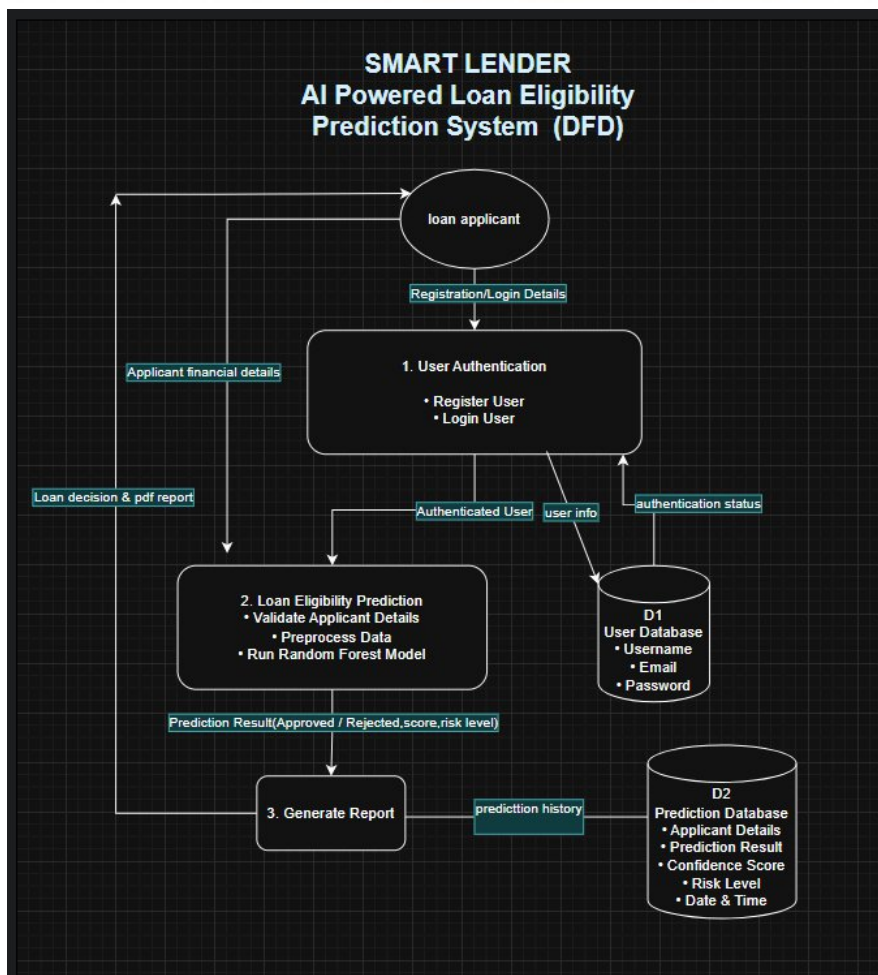
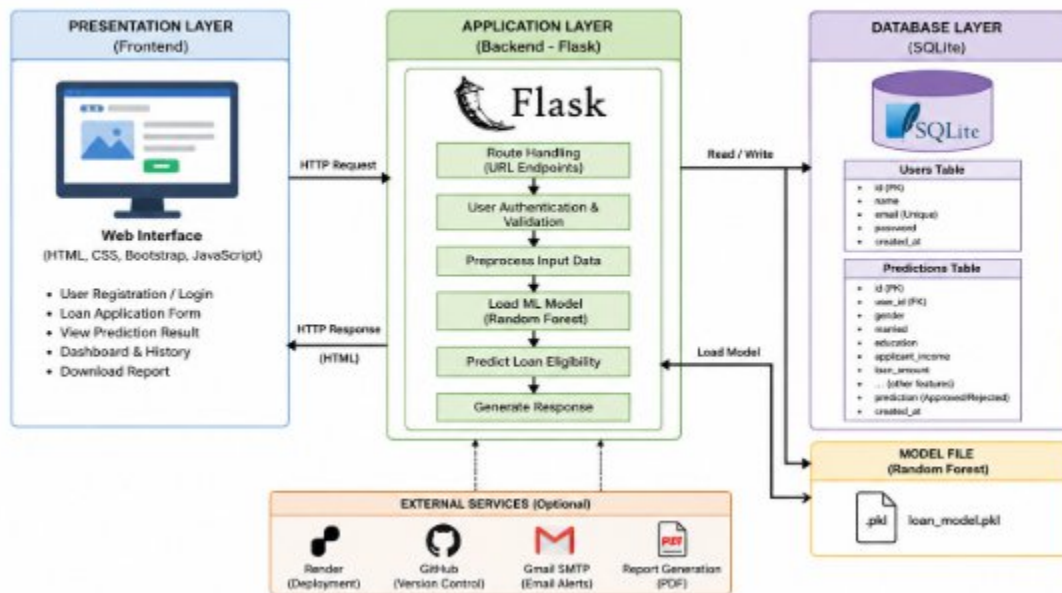
An applicant with low income and poor credit history submits a loan request. The Random Forest model predicts that the loan should be rejected, helping reduce financial risk.

### Scenario 3 – Faster Decision Making

A financial institution receives hundreds of loan applications every day. Smart Lender automates the loan evaluation process, reducing manual effort and improving processing speed.

## Technical Architecture:

### Technical Architecture:



## Description

Smart Lender follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

The Presentation Layer provides a responsive user interface developed using HTML, CSS,

Bootstrap, and JavaScript. Users can register, log in, and submit loan application details through an interactive web interface.

The Application Layer is developed using the Flask framework. It handles user authentication, validates loan application data, loads the trained Random Forest model, predicts loan eligibility, and generates the final prediction.

The Database Layer uses SQLite to securely store user credentials and loan prediction records. All prediction results are saved for future reference and analysis.

This architecture ensures scalability, security, faster response time, and efficient loan processing.

## Pre-requisites

- Python 3.10 or above
- Flask Framework
- Machine Learning using Scikit-Learn
- Pandas
- NumPy
- Pickle
- HTML
- CSS
- Bootstrap
- JavaScript
- SQLite Database
- VS Code
- Git & GitHub
- Render Deployment

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# Project Workflow

## Milestone 1: Model Selection and Architecture

### Activity 1.1: Dataset Collection

Collect the loan prediction dataset containing applicant information and loan status.

loan\_prediction .xlsx ☆ ☁ ☁

File Edit View Insert Format Data Tools Help

Q Menus 100% View only

|    | A        | B      | C       | D          | E              | F             | G               | H                 | I          | J             | K              | L             | M           |
|----|----------|--------|---------|------------|----------------|---------------|-----------------|-------------------|------------|---------------|----------------|---------------|-------------|
| 1  | Loan_ID  | Gender | Married | Dependents | Education      | Self_Employed | ApplicantIncome | CoapplicantIncome | LoanAmount | Loan_Amount_T | Credit_History | Property_Area | Loan_Status |
| 2  | LP001002 | Male   | No      |            | 0 Graduate     | No            | 5849            | 0                 |            | 360           | 1              | Urban         | Y           |
| 3  | LP001003 | Male   | Yes     |            | 1 Graduate     | No            | 4583            | 1508              | 128        | 360           | 1              | Rural         | N           |
| 4  | LP001005 | Male   | Yes     |            | 0 Graduate     | Yes           | 3000            | 0                 | 66         | 360           | 1              | Urban         | Y           |
| 5  | LP001006 | Male   | Yes     |            | 0 Not Graduate | No            | 2583            | 2358              | 120        | 360           | 1              | Urban         | Y           |
| 6  | LP001008 | Male   | No      |            | 0 Graduate     | No            | 6000            | 0                 | 141        | 360           | 1              | Urban         | Y           |
| 7  | LP001011 | Male   | Yes     |            | 2 Graduate     | Yes           | 5417            | 4196              | 267        | 360           | 1              | Urban         | Y           |
| 8  | LP001013 | Male   | Yes     |            | 0 Not Graduate | No            | 2333            | 1516              | 95         | 360           | 1              | Urban         | Y           |
| 9  | LP001014 | Male   | Yes     | 3+         | Graduate       | No            | 3036            | 2504              | 158        | 360           | 0              | Semiurban     | N           |
| 10 | LP001018 | Male   | Yes     |            | 2 Graduate     | No            | 4006            | 1526              | 168        | 360           | 1              | Urban         | Y           |
| 11 | LP001020 | Male   | Yes     |            | 1 Graduate     | No            | 12841           | 10968             | 349        | 360           | 1              | Semiurban     | N           |
| 12 | LP001024 | Male   | Yes     |            | 2 Graduate     | No            | 3200            | 700               | 70         | 360           | 1              | Urban         | Y           |
| 13 | LP001027 | Male   | Yes     |            | 2 Graduate     |               | 2500            | 1840              | 109        | 360           | 1              | Urban         | Y           |
| 14 | LP001028 | Male   | Yes     |            | 2 Graduate     | No            | 3073            | 8106              | 200        | 360           | 1              | Urban         | Y           |
| 15 | LP001029 | Male   | No      |            | 0 Graduate     | No            | 1853            | 2840              | 114        | 360           | 1              | Rural         | N           |
| 16 | LP001030 | Male   | Yes     |            | 2 Graduate     | No            | 1299            | 1086              | 17         | 120           | 1              | Urban         | Y           |
| 17 | LP001032 | Male   | No      |            | 0 Graduate     | No            | 4950            | 0                 | 125        | 360           | 1              | Urban         | Y           |
| 18 | LP001034 | Male   | No      |            | 1 Not Graduate | No            | 3596            | 0                 | 100        | 240           |                | Urban         | Y           |
| 19 | LP001036 | Female | No      |            | 0 Graduate     | No            | 3510            | 0                 | 76         | 360           | 0              | Urban         | N           |
| 20 | LP001038 | Male   | Yes     |            | 0 Not Graduate | No            | 4887            | 0                 | 133        | 360           | 1              | Rural         | N           |
| 21 | LP001041 | Male   | Yes     |            | 0 Graduate     |               | 2600            | 3500              | 115        |               | 1              | Urban         | Y           |
| 22 | LP001043 | Male   | Yes     |            | 0 Not Graduate | No            | 7660            | 0                 | 104        | 360           | 0              | Urban         | N           |
| 23 | LP001046 | Male   | Yes     |            | 1 Graduate     | No            | 5955            | 5625              | 315        | 360           | 1              | Urban         | Y           |
| 24 | LP001047 | Male   | Yes     |            | 0 Not Graduate | No            | 2600            | 1911              | 116        | 360           | 0              | Semiurban     | N           |
| 25 | LP001050 |        | Yes     |            | 2 Not Graduate | No            | 3365            | 1917              | 112        | 360           | 0              | Rural         | N           |
| 26 | LP001052 | Male   | Yes     |            | 1 Graduate     |               | 3717            | 2925              | 151        | 360           |                | Semiurban     | N           |
| 27 | LP001066 | Male   | Yes     |            | 0 Graduate     | Yes           | 9560            | 0                 | 191        | 360           | 1              | Semiurban     | Y           |
| 28 | LP001068 | Male   | Yes     |            | 0 Graduate     | No            | 2799            | 2253              | 122        | 360           | 1              | Semiurban     | Y           |
| 29 | LP001073 | Male   | Yes     |            | 2 Not Graduate | No            | 4226            | 1040              | 110        | 360           | 1              | Urban         | Y           |
| 30 | LP001086 | Male   | No      |            | 0 Not Graduate | No            | 1442            | 0                 | 35         | 360           | 1              | Urban         | N           |
| 31 | LP001087 | Female | No      |            | 2 Graduate     |               | 3750            | 2083              | 120        | 360           | 1              | Semiurban     | Y           |
| 32 | LP001091 | Male   | Yes     |            | 1 Graduate     |               | 4166            | 3369              | 201        | 360           |                | Urban         | N           |
| 33 | LP001095 | Male   | No      |            | 0 Graduate     | No            | 3167            | 0                 | 74         | 360           | 1              | Urban         | N           |
| 34 | LP001097 | Male   | No      |            | 1 Graduate     | Yes           | 4692            | 0                 | 106        | 360           | 1              | Rural         | N           |
| 35 | LP001098 | Male   | Yes     |            | 0 Graduate     | No            | 3500            | 1667              | 114        | 360           | 1              | Semiurban     | Y           |

### Activity 1.2: Data Preprocessing

Handle missing values, encode categorical variables, normalize data, and prepare the dataset for model training.

```
data.isnull().sum()
```

|                   |    |
|-------------------|----|
| Loan_ID           | 0  |
| Gender            | 13 |
| Married           | 3  |
| Dependents        | 15 |
| Education         | 0  |
| Self_Employed     | 32 |
| ApplicantIncome   | 0  |
| CoapplicantIncome | 0  |
| LoanAmount        | 22 |
| Loan_Amount_Term  | 14 |
| Credit_History    | 50 |
| Property_Area     | 0  |
| Loan_Status       | 0  |
| dtype: int64      |    |

```
data.describe()
```

|       | ApplicantIncome | CoapplicantIncome | LoanAmount | Loan_Amount_Term | Credit_History |
|-------|-----------------|-------------------|------------|------------------|----------------|
| count | 614.000000      | 614.000000        | 592.000000 | 600.000000       | 564.000000     |
| mean  | 5403.459283     | 1621.245798       | 146.412162 | 342.000000       | 0.842199       |
| std   | 6109.041673     | 2926.248369       | 85.587325  | 65.12041         | 0.364878       |
| min   | 150.000000      | 0.000000          | 9.000000   | 12.000000        | 0.000000       |
| 25%   | 2877.500000     | 0.000000          | 100.000000 | 360.000000       | 1.000000       |
| 50%   | 3812.500000     | 1188.500000       | 128.000000 | 360.000000       | 1.000000       |
| 75%   | 5795.000000     | 2297.250000       | 168.000000 | 360.000000       | 1.000000       |

## Activity 1.3: Model Training

Train multiple machine learning models such as Decision Tree, KNN, Random Forest, and XGBoost.

0
Profile
requirements.txt Flask
runtime.txt
loan\_prediction.ipynb X
style.css
Smart Lender
predict.html

loan\_prediction.ipynb >
data.isnull().sum()

Generate

+ Code

+ Markdown

Run All

...

|   |      |      |      |    |
|---|------|------|------|----|
| 0 | 0.57 | 0.56 | 0.56 | 43 |
| 1 | 0.77 | 0.78 | 0.77 | 80 |

|              |      |      |     |
|--------------|------|------|-----|
| accuracy     |      | 0.70 | 123 |
| macro avg    | 0.67 | 0.67 | 123 |
| weighted avg | 0.70 | 0.70 | 123 |

```
[38] rf_model = RandomForestClassifier(random_state=42)
```

```
[39] rf_model.fit(X_train, y_train)
```

RandomForestClassifier

Parameters

Fitted attributes

```
[40] rf_pred = rf_model.predict(X_test)
```

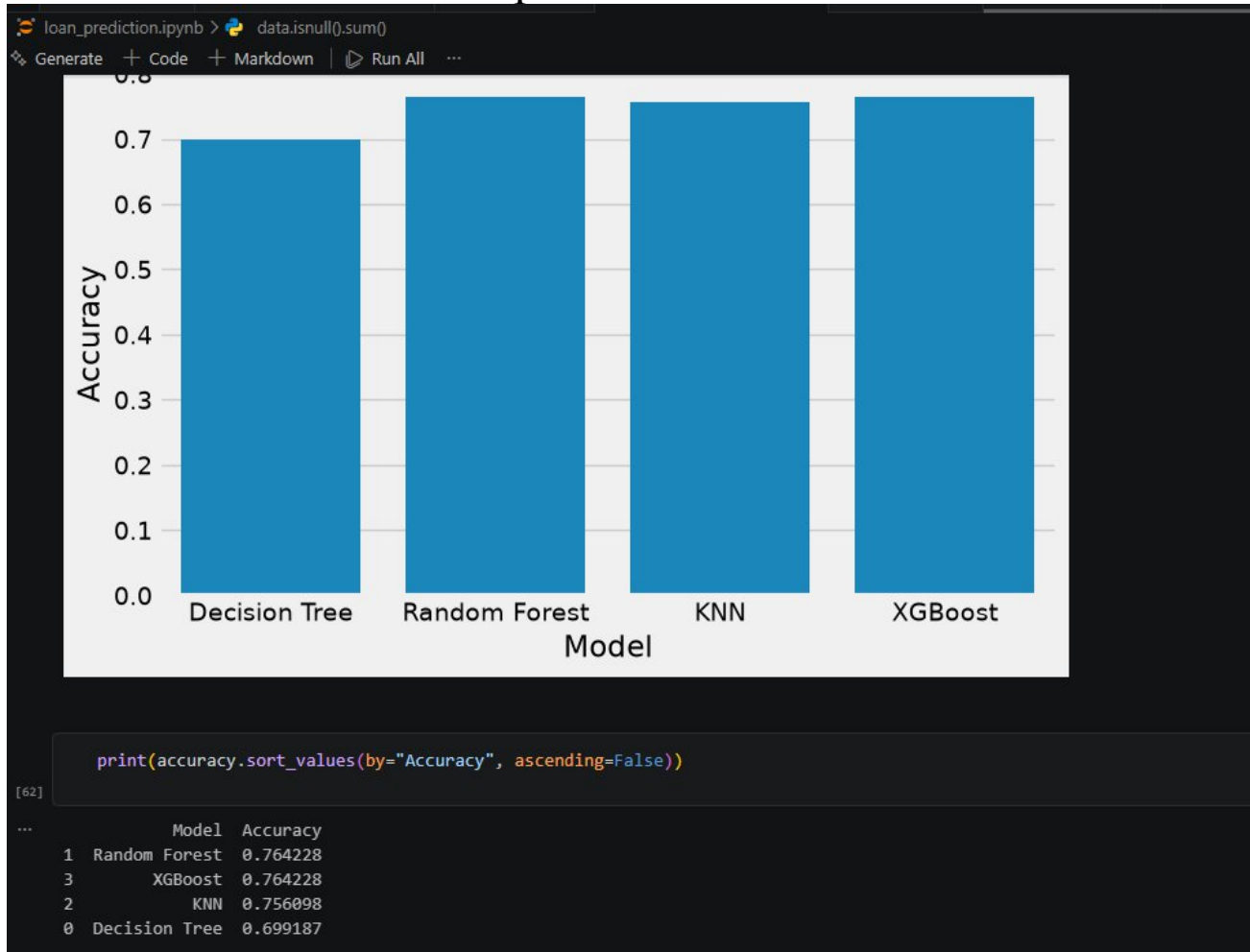
```
[41] rf_accuracy = accuracy_score(y_test, rf_pred)
```

```
print("Random Forest Accuracy :", rf_accuracy)
```

```
Random Forest Accuracy : 0.7642276422764228
```

## Activity 1.4: Model Selection

Compare model performance using Accuracy Score and select Random Forest as the final prediction model.



## Milestone 2: Core Functionalities Development

### Activity 2.1

Develop User Registration and Login Module.

### Activity 2.2

Implement Loan Prediction Module.

### Activity 2.3

Validate applicant information before prediction.

### Activity 2.4

Store prediction results in SQLite Database.

### Activity 2.5

Generate prediction report.





## Smart Lender

### Loan Eligibility Assessment Report

Powered by Artificial Intelligence & Machine Learning

# ✕ LOAN REJECTED

Better luck next time

#### Applicant Details

**Name :** Madhan

**Income :** ₹ 10000.0

**Loan Amount :** ₹ 500000.0

**Credit History :** Good

**Property Area :** Urban

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#### Applicant Summary

- ✓ Income Verified
- ✓ Credit History Verified
- ✓ AI Risk Analysis Completed
- ✓ Loan Eligibility Checked

#### Prediction Summary

**Prediction Result**

Loan Rejected ✕

**Confidence Score**

52.55%

**Risk Assessment**

HIGH RISK

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#### AI Model Details

Algorithm : Random Forest Classifier

Training Accuracy : 94.7%

Testing Accuracy : 81.1%

Prediction Engine : Smart Lender AI v3

**Date :** 29 June 2026

**Time :** 10:10 AM

+ Predict Again

Dashboard

Print

Download PDF



Smart Lender

AI Powered Loan Eligibility Prediction System

Generated on 29 June 2026 at 10:10 AM

Developed using Flask, SQLite, Scikit-learn and Random Forest Classifier

## Milestone 3: Flask Application Development

### Activity 3.1

Create Flask routes for

- Home Page
- Login
- Register
- Loan Prediction
- Dashboard
- Logout

```

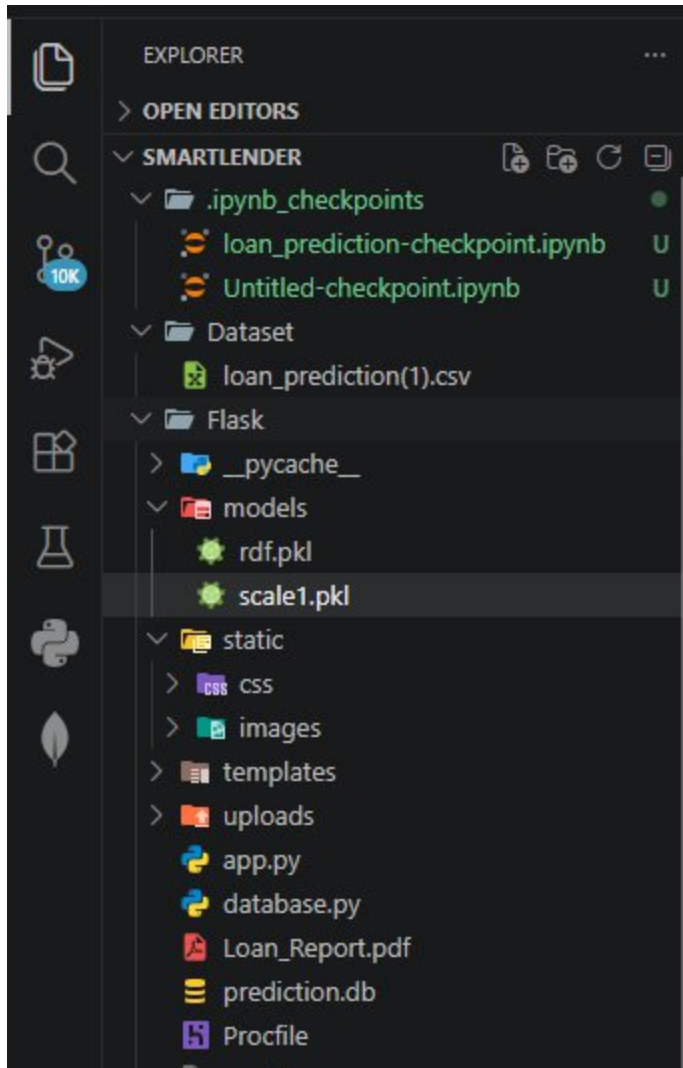
app.py x requirements.txt \ .gitignore U Smart Lender | AI Loan Eligibi... history.html Smart Lende
Flask > app.py > download_pdf
9 from reportlab.lib.styles import getSampleStyleSheet
10 from reportlab.pdfbase import pdfmetrics
11 import os
12 from reportlab.lib.enums import TA_CENTER
13 from reportlab.lib.styles import ParagraphStyle
14 app = Flask(__name__)
15 app.secret_key = "smart_lender_secret_key"
16
17 # Load model and scaler
18 model = pickle.load(open("models/rdf.pkl", "rb"))
19 scaler = pickle.load(open("models/scale1.pkl", "rb"))
20
21 create_database()
22 create_users_table()
23
24 @app.route("/register")
25 def register():
26
27     return render_template("register.html")
28
29 @app.route("/login")
30 def login():
31
32     return render_template("login.html")
33
34 @app.route("/logout")
35
36 def logout():
37
38     session.clear()
39
40     return redirect("/")
41
42 @app.route("/")
43 def home():
44     return render_template("home.html")
45 @app.route("/dashboard")
46 def dashboard():

```

## Activity 3.2

Load the trained Random Forest model using Pickle.





### Activity 3.3

Accept applicant information from HTML form.

### Activity 3.4

Predict loan eligibility using the trained model.

### Activity 3.5

Store prediction details inside SQLite Database.

app.py | prediction.db | requirements.txt | .gitignore | Smart Lender | AI Loan Eligibi... | history.html | Smart Lender | AI Loan Eligibi...

Flask > prediction.db

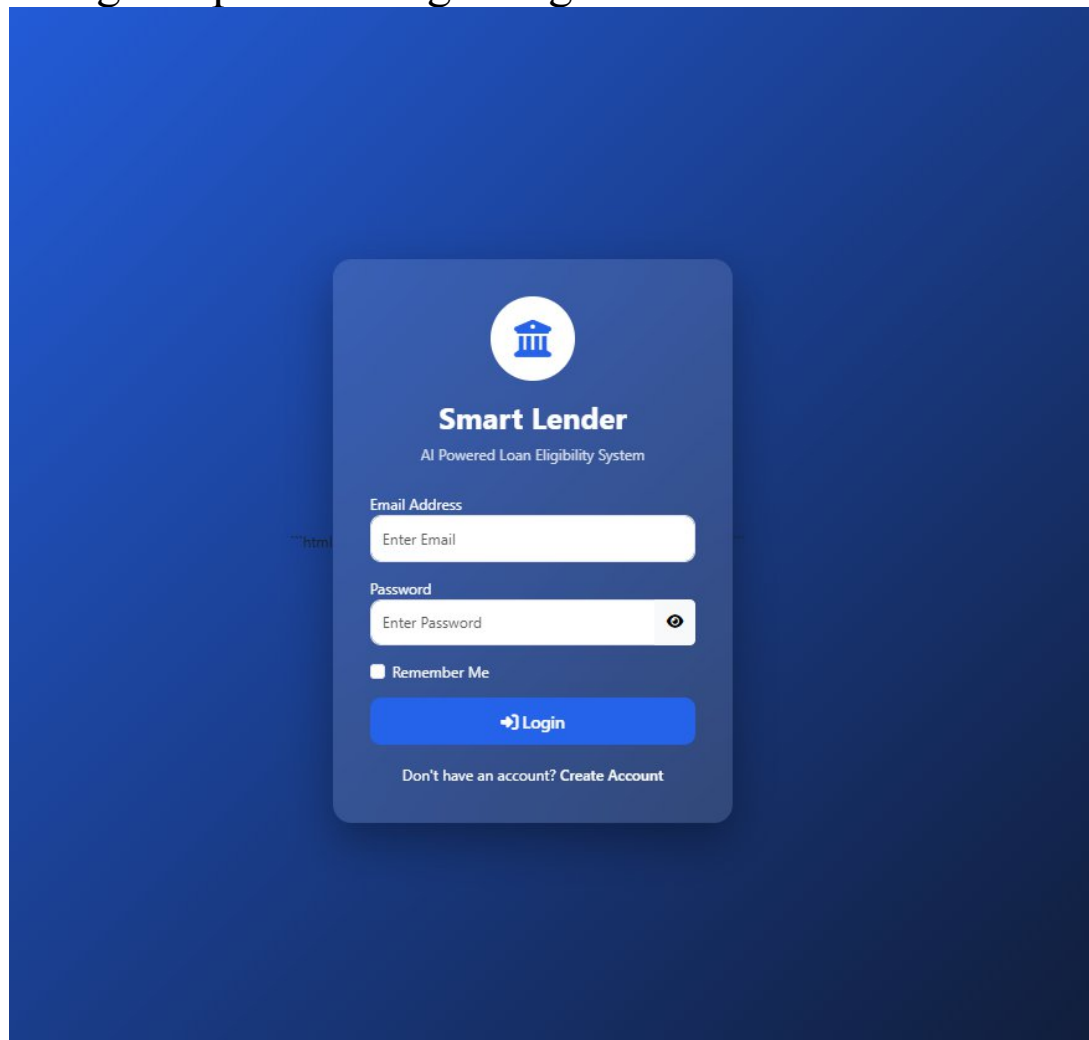
Tables: Rows: 10

|    | id | applicant... | gender | married | depende... | education | self_emp... | applicant... | coapplic... |
|----|----|--------------|--------|---------|------------|-----------|-------------|--------------|-------------|
| 1  | 1  | pavan        | 1.0    | 1.0     | 1.0        | 1.0       | 1.0         | 2000         | 30000       |
| 2  | 2  | king         | 0.0    | 0.0     | 2.0        | 1.0       | 1.0         | 10000        | 20000       |
| 3  | 3  | ravi         | 1.0    | 1.0     | 0.0        | 1.0       | 1.0         | 10000        | 20000       |
| 4  | 4  | ravi         | 1.0    | 1.0     | 0.0        | 1.0       | 1.0         | 10000        | 20000       |
| 5  | 5  | siddu        | 1.0    | 1.0     | 1.0        | 1.0       | 1.0         | 10000        | 200000      |
| 6  | 6  | pavan        | 1.0    | 1.0     | 0.0        | 1.0       | 1.0         | 122000       | 10000       |
| 7  | 7  | king         | 0.0    | 0.0     | 1.0        | 0.0       | 1.0         | 2000         | 10000       |
| 8  | 8  | king         | 0.0    | 0.0     | 1.0        | 0.0       | 1.0         | 2000         | 10000       |
| 9  | 9  | siddu        | 1.0    | 1.0     | 0.0        | 1.0       | 1.0         | 10000        | 2000        |
| 10 | 10 | sagar        | 1.0    | 1.0     | 2.0        | 1.0       | 1.0         | 50000        | 100000      |
| +  | 11 |              |        |         |            |           |             |              |             |

## Milestone 4: Frontend Development

### Activity 4.1

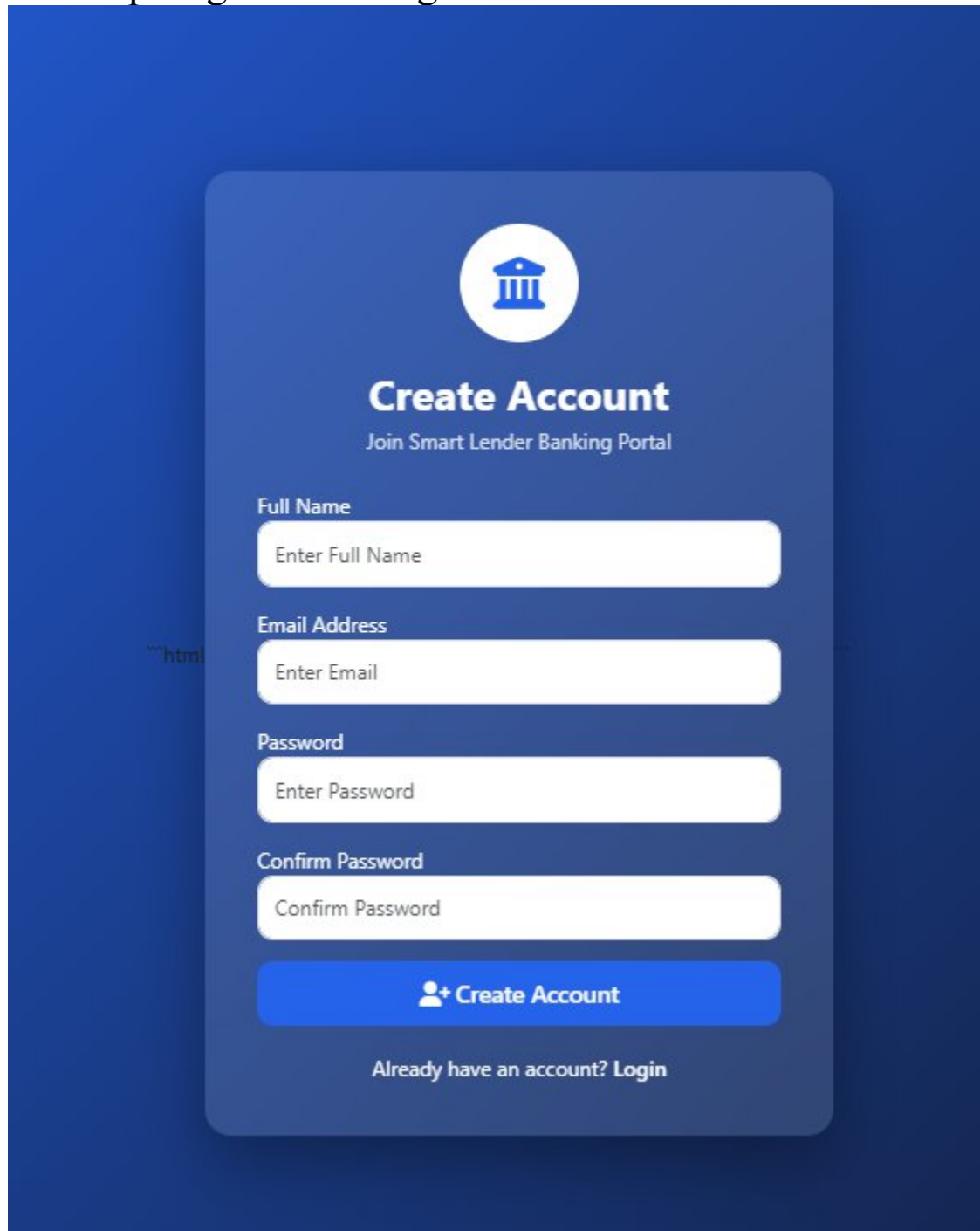
Design responsive Login Page.



The login page features a dark blue background with a central white login card. The card contains the Smart Lender logo (a blue building icon in a circle), the text "Smart Lender" and "AI Powered Loan Eligibility System". Below this are input fields for "Email Address" and "Password", each with a placeholder "Enter Email" and "Enter Password" respectively. A "Remember Me" checkbox is located below the password field. A blue "Login" button with a right-pointing arrow is positioned below the checkbox. At the bottom of the card, there is a link that says "Don't have an account? Create Account".

## Activity 4.2

Develop Registration Page.



The image shows a registration form titled "Create Account" for the "Smart Lender Banking Portal". The form is set against a dark blue background. At the top of the form is a circular icon containing a blue building with columns. Below the icon, the title "Create Account" is displayed in a large, bold, white font, followed by the subtitle "Join Smart Lender Banking Portal" in a smaller, lighter blue font. The form contains four input fields, each with a label above it: "Full Name" (with placeholder text "Enter Full Name"), "Email Address" (with placeholder text "Enter Email"), "Password" (with placeholder text "Enter Password"), and "Confirm Password" (with placeholder text "Confirm Password"). Below these fields is a prominent blue button with a white user icon and the text "Create Account". At the bottom of the form, there is a link that says "Already have an account? Login".

## Activity 4.3

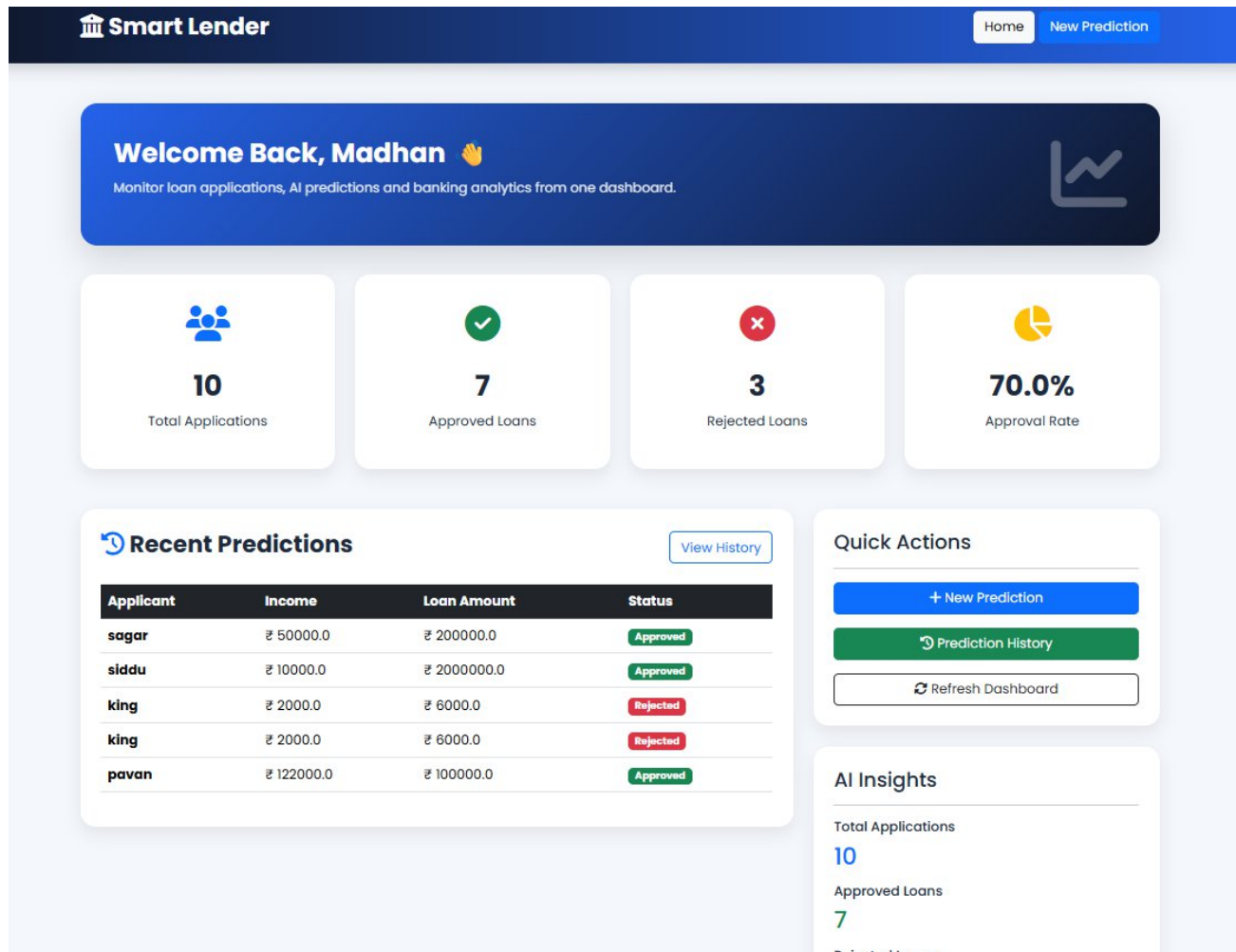
Develop Loan Prediction Form.

## Activity 4.4

Display Prediction Results.

## Activity 4.5

Develop Dashboard for Prediction History.



## Milestone 5: Deployment

### Activity 5.1

Create Virtual Environment

```
python -m venv venv
```

Activate Environment

```
venv\Scripts\activate
```

Install Dependencies

```
pip install -r requirements.txt
```

### Activity 5.2

Run Flask Application

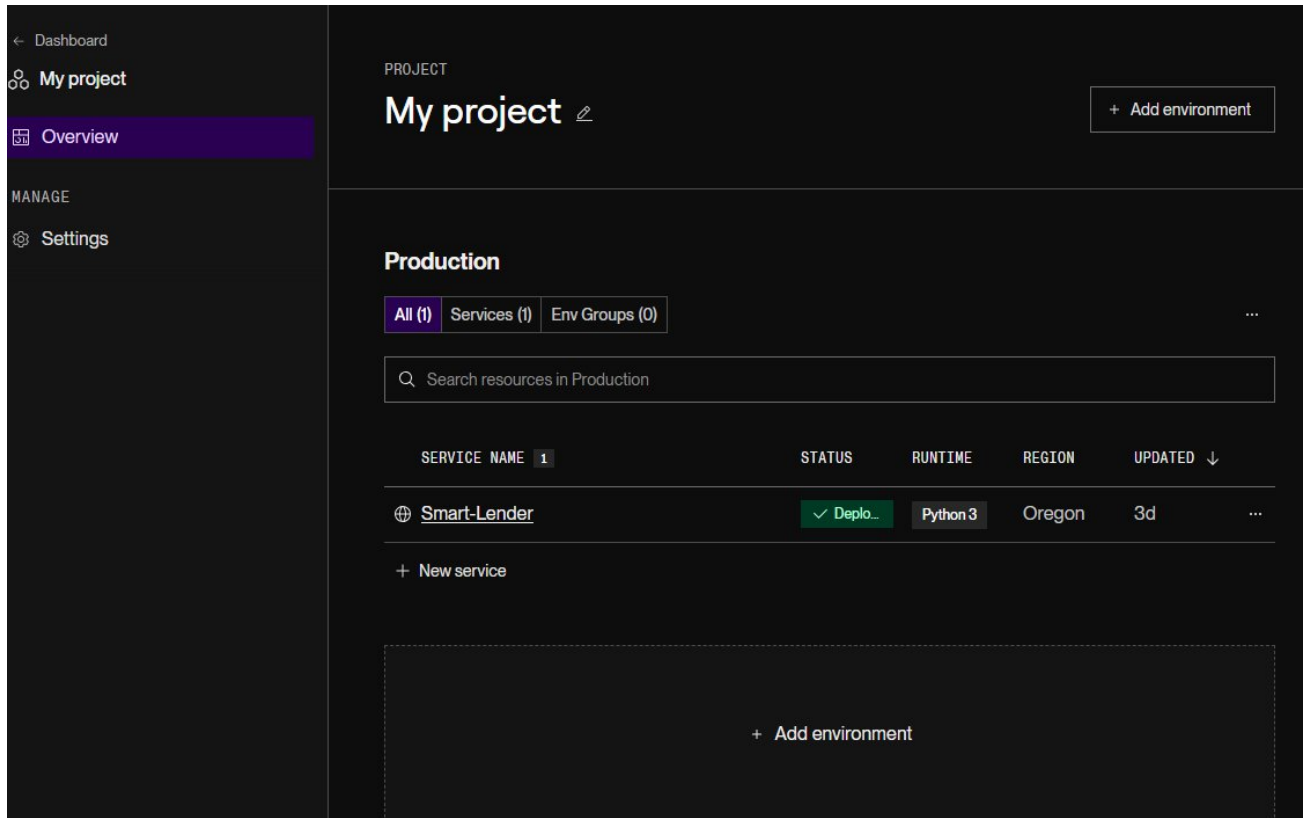
```
python app.py
```

### Activity 5.3

Push Source Code to GitHub.

### Activity 5.4

Deploy the application on Render.



The screenshot shows the 'My project' dashboard in the SmartBridge interface. The left sidebar contains navigation links: 'Dashboard', 'My project', 'Overview' (highlighted), 'MANAGE', and 'Settings'. The main content area is titled 'PROJECT My project' and includes a '+ Add environment' button. Below this, the 'Production' environment is selected, showing tabs for 'All (1)', 'Services (1)', and 'Env Groups (0)'. A search bar is present with the text 'Search resources in Production'. A table lists the services in the Production environment:

| SERVICE NAME | STATUS      | RUNTIME | REGION | UPDATED |
|--------------|-------------|---------|--------|---------|
| Smart-Lender | ✓ Deploy... | Python3 | Oregon | 3d      |

Below the table, there is a '+ New service' link and a large dashed box containing a '+ Add environment' button.

## Activity 5.5

Test the deployed application.

<https://smart-lender-gmxw.onrender.com/>

## **Conclusion:**

Smart Lender is an AI-powered Loan Eligibility Prediction System that assists banks and financial institutions in making faster and more accurate loan approval decisions. By leveraging the Random Forest Machine Learning algorithm, the application analyzes applicant information and predicts loan eligibility with high accuracy.

The project integrates Machine Learning, Flask, HTML, CSS, Bootstrap, JavaScript, and SQLite into a complete web-based solution. It reduces manual effort, speeds up loan processing, and minimizes financial risk. The system is scalable, user-friendly, and suitable for real-world banking applications.

Future enhancements may include cloud database integration, support for multiple machine learning models, real-time analytics dashboards, document verification, and deployment using advanced cloud platforms to further improve performance and usability.